

# Mohammad Alvi Refat

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## EDUCATION

**The Graduate Center, City University of New York**

*Sep 2025*

M.S. in Astrophysics

*Advisor:* Dr. Lucy Lu

*Thesis:* Starspot Inference Using Light Curve Inversion Techniques

**CUNY Baccalaureate for Unique and Interdisciplinary Studies**

*Jun 2022*

B.S. in Computational Astrophysics, *cum laude*

## RESEARCH INTERESTS

Computational physics; numerical modeling and simulation; scientific computing; statistical inference and inverse problems; applications across astrophysics, materials physics, biophysics, and complex systems.

## RESEARCH EXPERIENCE

**Starspot Inference with Light Curve Inversion Techniques** [\[Thesis\]](#)

*American Museum of Natural History*

*Aug 2023 – Sep 2025*

*M.S. Thesis Researcher | Advisor: Dr. Lucy Lu*

- Implemented forward- and inverse-modeling workflows with **fleck** and **starry** to analyze 1,000 synthetic, evolving starspot light curves generated with **butterfly**.
- Evaluated recovery of stellar rotation, activity indicators, and surface structure across variations in photometric noise, viewing inclination, and spot configuration.
- Recovered rotation and statistical activity indicators in the tested light curves, while finding that individual starspot locations and surface maps were not uniquely constrained.

**Towards Mapping Brown Dwarf and Giant Exoplanet Atmospheres** [\[Code\]](#)

*American Museum of Natural History*

*May 2021 – Aug 2023*

*BDNYC Researcher | Advisor: Dr. Johanna Vos*

- Fit synthetic light curves derived from three-dimensional atmospheric simulations and 20-hour Spitzer observations of two brown dwarfs with **starry** spherical-harmonic surface maps.
- Evaluated model complexity, identifying spherical-harmonic degree  $l = 5$  as sufficient to reproduce the relevant light-curve structure.
- Identified atmospheric evolution as a limitation of static-map fits, motivating rotation-by-rotation mapping of changing cloud structures.

**Chemodynamically Characterizing the Jhelum Stellar Stream with APOGEE-2** [\[Code\]](#) [\[Paper\]](#)

*Sloan Digital Sky Survey Faculty and Student Team*

*May 2020 – May 2021*

*SDSS FAST Student Researcher | Advisor: Dr. Allyson Sheffield*

- Analyzed APOGEE-2 chemical abundances for stars observed toward the Jhelum stream, comparing elemental abundance ratios with previously identified members and other Milky Way populations.
- Produced abundance–metallicity comparisons that helped assess the chemical properties and possible progenitor of the stream.
- Coauthored a peer-reviewed article in *The Astrophysical Journal* reporting the collaboration’s chemodynamical analysis.

## Looking at Star Formation Through Chemistry

American Museum of Natural History

Aug 2018 – May 2020

AstroCom NYC Scholar | Advisor: Dr. Allyson Sheffield

- Processed and normalized high-resolution echelle spectra with **SMHr**, measured absorption-line equivalent widths, and iteratively fitted stellar atmospheric parameters and elemental abundances for chemical-tagging analysis.
- Validated the workflow against published Arcturus measurements, recovering effective temperature within 50 K and metallicity within 0.07 dex of the reference values.

## Two-Point Statistics in the Star-Forming Interstellar Medium [\[Code\]](#)

Flatiron Institute, Center for Computational Astrophysics

Jun 2018 – Aug 2018

AstroCom NYC Scholar | Advisor: Dr. Chang-Goo Kim

- Developed Python routines to compute and visualize two-point metallicity correlations in TIGRESS magnetohydrodynamic simulations of the star-forming interstellar medium.
- Measured a characteristic metallicity half-correlation length of  $120 \pm 25$  pc and estimated a turbulent metal-mixing timescale of  $\sim 10$  Myr.

## Staying in Science

American Museum of Natural History

Aug 2017 – Sep 2019

Research Contributor, NSF-Funded Study | Advisor: Dr. Preeti Gupta

- Prepared research memos and contributed participant information used to develop social-network maps of students' STEM education and career pathways.
- Reviewed survey data to examine STEM persistence among students from historically marginalized backgrounds who participated in science research mentoring programs.

## PUBLICATIONS

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### Peer-Reviewed Publications

Sheffield, A. A., A. Z. Subrahimovic, **M. Refat**, et al. (2021). “Chemodynamically Characterizing the Jhelum Stellar Stream with APOGEE-2”. *The Astrophysical Journal* 913, Article 39. [DOI](#).

### Thesis

**Refat, M.** (2025). “[Starspot Inference Using Light Curve Inversion Techniques](#)”. M.S. Thesis. CUNY Graduate Center.

## ACADEMIC PRESENTATIONS

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### Talks

|                                                                                                                                                     |      |
|-----------------------------------------------------------------------------------------------------------------------------------------------------|------|
| “Star Spot Inference Using Light Curve Inversion Techniques”<br><i>CUNY Masters Graduation</i> — Simons Foundation                                  | 2025 |
| “Towards Mapping Brown Dwarf and Giant Exoplanet Atmospheres”<br><i>240th American Astronomical Society Meeting</i> — Pasadena, CA                  | 2022 |
| “Towards Mapping Brown Dwarf and Giant Exoplanet Atmospheres”<br><i>Center for Computational Astrophysics Symposium</i> — Simons Foundation         | 2021 |
| “Chemodynamically Characterizing the Jhelum Stellar Stream with APOGEE-2”<br><i>American Museum of Natural History REU Symposium</i> — New York, NY | 2020 |

|                                                                                |      |
|--------------------------------------------------------------------------------|------|
| <b>“Where Were Stars Born in the Milky Way?”</b>                               | 2019 |
| <i>American Museum of Natural History REU Symposium</i> — New York, NY         |      |
| <b>“Two-Point Statistics in the Star-Forming ISM”</b>                          | 2018 |
| <i>Center for Computational Astrophysics REU Symposium</i> — Simons Foundation |      |

## Posters

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|----------------------------------------------------------------------------------|------|
| <b>“Starspot Inference with Light Curve Inversion Techniques”</b>                | 2025 |
| <i>245th American Astronomical Society Meeting</i> — National Harbor, MD         |      |
| <b>“Star Spots with Starry”</b>                                                  | 2024 |
| <i>Cool Stars 22</i> — University of California San Diego                        |      |
| <b>“Towards Mapping Brown Dwarf and Giant Exoplanet Atmospheres”</b>             | 2023 |
| <i>Center for Computational Astrophysics Gotham Fest</i> — Simons Foundation     |      |
| <b>“Towards Mapping Brown Dwarf and Giant Exoplanet Atmospheres”</b>             | 2023 |
| <i>241st American Astronomical Society Meeting</i> — Seattle, WA                 |      |
| <b>“Towards Mapping Brown Dwarf and Giant Exoplanet Atmospheres”</b>             | 2021 |
| <i>Center for Computational Astrophysics Gotham Fest</i> — Simons Foundation     |      |
| <b>“Chemodynamically Characterizing the Jhelum Stellar Stream with APOGEE-2”</b> | 2021 |
| <i>237th American Astronomical Society Meeting</i> — Virtual                     |      |
| <b>“Where Were Stars Born in the Milky Way?”</b>                                 | 2019 |
| <i>Center for Computational Astrophysics Gotham Fest</i> — New York University   |      |

## PROFESSIONAL APPOINTMENTS

|                                                                                                                                                                        |                            |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| <b>Adjunct Lecturer</b>                                                                                                                                                | <i>Aug 2025 – May 2026</i> |
| <i>Baruch College, City University of New York</i>                                                                                                                     |                            |
| – Delivered introductory physics lecture and laboratory instruction for PHYS 2002, PHYS 2003, and PHYS 3001.                                                           |                            |
| – Designed lectures, laboratory exercises, and assessments emphasizing experimental methods, computational problem solving, data analysis, and uncertainty evaluation. |                            |

## COURSES TAUGHT

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|----------------------------------------------------|------------------------|
| <b>PHYS 2002 – Introductory Physics Lecture</b>    | Fall 2025, Spring 2026 |
| <b>PHYS 2003 – Introductory Physics Laboratory</b> | Fall 2025              |
| <b>PHYS 3001 – Physics II Laboratory</b>           | Spring 2026            |

## TECHNICAL SKILLS

|                              |                                                                                                                     |
|------------------------------|---------------------------------------------------------------------------------------------------------------------|
| <b>Programming Languages</b> | Python, MATLAB, Bash, Java                                                                                          |
| <b>Computational Methods</b> | Inverse problems, Bayesian inference, numerical optimization, time-series and spectral analysis, spatial statistics |
| <b>Scientific Computing</b>  | NumPy, SciPy, Pandas, Astropy, Lightkurve, Astroquery                                                               |
| <b>Visualization</b>         | Matplotlib, Glue, Manim                                                                                             |
| <b>Research Tools</b>        | Git, Linux, Jupyter Notebook, L <sup>A</sup> T <sub>E</sub> X                                                       |

## OUTREACH

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|-------------------------------------------------------------------------|------------|
| <b>Science Research Mentoring Program Meeting — Speaker</b>             | 2021       |
| <b>Science Research Mentoring Program Family Orientation — Panelist</b> | 2019, 2020 |